

WASP-12b

R-1847

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160130 · created 2026-08-03T07:53:01+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457418.70214 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.028
Transit depth (Rp/Rs)^2	1.45 +/- 0.68 [%]
Orbital Inclination (inc)	86.71 +/- 2.62
Airmass coefficient 1 (a1)	1.064 +/- 0.019
Airmass coefficient 2 (a2)	-0.059 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#1 - [491, 393]
Optimal Aperture	3.55
Optimal Annulus	7.7
Transit Duration (day)	0.1287 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.121 ± 0.028 vs 0.1178 (deviation 0.11σ)
Duration fit vs published	0.1287 d vs 0.125 d (deviation 0.88σ)
Mid-time O–C	1571.0 min
Depth SNR	4.3
Residual scatter	1.77 %

Light curve

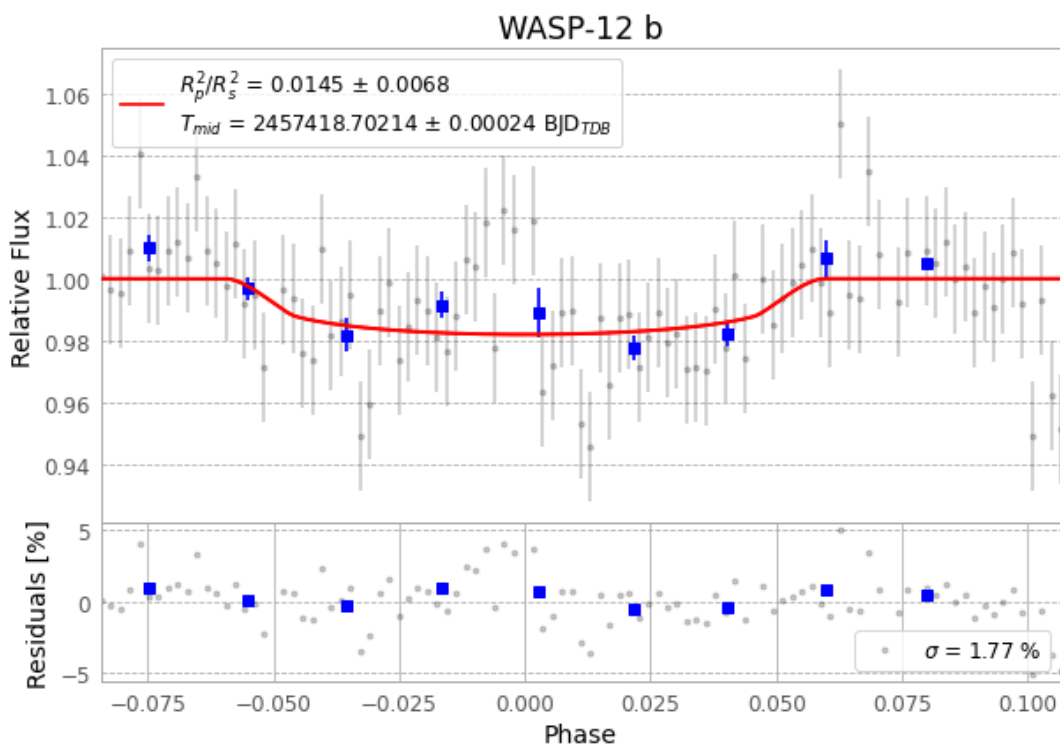


Figure 1 — FinalLightCurve_WASP-12 b_2016-01-30.png (SHA-256 221dcb0658002f9f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 213], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0873bb581091e9d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12160131023012.FITS → WASP-12160131073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160130.log (0e800b3f6b19...), FinalLightCurve_WASP-12 b_2016-01-30.png (221dcb065800...), FinalLightCurve_WASP-12 b_2016-01-30.csv (b16c1688da0f...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 07:53Z.